Vedic Maths

Formula & Shortcut Sheets

Sutra chart · tables · identities · aptitude formulas
Common mistakes · exam-day strategy · revision checklists

$97 \times 96 = 93|12$

$35^2 = 12|25$

$1000 - 357 = 643$

$43 \times 11 = 473$

Original practice material. Exam names are used only to describe the syllabus; no question is an official previous-year question.

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Part XV

Revision and Formula Sheets

Use these pages for last-minute revision. Each sheet fits a page or two, so you can print the separate **Formula Sheets** PDF and keep it on your desk.

Chapter 57: The Sutra Chart

#	Sutra	Meaning	Use it for	Example
1	Ekadhikena Purvena	One more than the previous	squares ending in 5; 1/19	$65^2 \to 6 \times 7 \mid 25 \to 4{,}225$
2	Nikhilam	All from 9, last from 10	$10^n - n$; near-base × and ÷	$1{,}000 - 357 \to 643$
3	Urdhva-Tiryag bhyam	Vertically and crosswise	any multiplication	$23 \times 41 \to 943$
4	Paravartya Yojayet	Transpose and adjust	divisors just above a base; equations	$1{,}234 \div 12 \to$ Q 102, R 10
5	Shunyam Saa myasamuccay e	Same total → zero	special equations	$(x + 7)(x + 9) = (x + 3)(x + 21)$: 7×9 and 3×21 are both $63 \to x = 0$
6	Anurupye Shunyamanyat	One in ratio, other zero	simultaneous equations	$2x + 3y = 6$, $7x + 9y = 18 \to x = 0$
7	Sankalana-Vya vakalanabhya m	Add and subtract	swapped coefficients	add and subtract the equations
8	Puranapurana bhyam	Completion	completing the square	$x^2 + 6x \to (x + 3)^2 - 9$
9	Chalana-Kalan abhyam	Differential	quadratic roots	$2x + b = \pm\sqrt{(b^2 - 4ac)}$
10	Yavadunam	By the deficiency	squares near a base	$96^2 \to 92 \mid 16 \to 9{,}216$
11	Vyashtisamans tih	Part and whole	splitting, averages	$49 \times 51 = 50^2 - 1$
12	Shesanyanken a Charamena	Remainders by last digit	decimals of 1/7, 1/13	$1/7 = 0.142857\ldots$
13	Sopantyadvay amantyam	Last + twice penultimate	special fraction equations	see Part II
14	Ekanyunena Purvena	One less than the previous	× 9, 99, 999	$47 \times 99 \to 46 \mid 53 \to 4{,}653$
15	Gunitasamucc ayah	Sum-product check	checking products	digit sums
16	Gunakasamuc cayah	Factors of the sum	checking factorisations	$x = 1$ check

Chapter 58: Formula Sheets

58.1 Multiplication shortcuts

Situation	Method
× 5, 25, 125	× 10/2, × 100/4, × 1000/8
× 9, 99, 999 (same number of digits)	(n − 1) | (complement of n)
× 11	write the outer digits, add neighbours in between
× 101	write the number twice (for 2 digits)
Both near a base	$(a - d_2) \mid d_1 \times d_2$, with signs
Same tens, units add to 10	tens × (tens + 1) | units × units
General 2-digit	right: vertical; middle: crosswise sum; left: vertical
Squares ending in 5	n(n + 1) | 25
Squares near 50	$(25 \pm d) \mid d^2$
Squares near 100	$(n \pm d) \mid d^2$
Any square	duplex method

58.2 Squares and cubes

n	n^2	n	n^2	n	n^2	n	n^2	n	n^2
1	1	7	49	13	169	19	361	25	625
2	4	8	64	14	196	20	400	26	676
3	9	9	81	15	225	21	441	27	729
4	16	10	100	16	256	22	484	28	784
5	25	11	121	17	289	23	529	29	841
6	36	12	144	18	324	24	576	30	900

n	n^3	n	n^3	n	n^3
1	1	6	216	11	1,331
2	8	7	343	12	1,728
3	27	8	512	13	2,197
4	64	9	729	14	2,744
5	125	10	1,000	15	3,375

58.3 Fractions and percentages

Fraction	Percentage	Decimal
1/2	50%	0.5
1/3	33 1/3%	0.3333… (recurring)
2/3	66 2/3%	0.6667… (recurring)
1/4	25%	0.25

Fraction	Percentage	Decimal
3/4	75%	0.75
1/5	20%	0.2
2/5	40%	0.4
3/5	60%	0.6
4/5	80%	0.8
1/6	16 2/3%	0.1667… (recurring)
5/6	83 1/3%	0.8333… (recurring)
1/7	14 2/7%	0.1429… (recurring)
1/8	12 1/2%	0.125
3/8	37 1/2%	0.375
5/8	62 1/2%	0.625
7/8	87 1/2%	0.875
1/9	11 1/9%	0.1111… (recurring)
1/10	10%	0.1
1/11	9 1/11%	0.0909… (recurring)
1/12	8 1/3%	0.0833… (recurring)
1/15	6 2/3%	0.0667… (recurring)
1/16	6 1/4%	0.0625
1/20	5%	0.05
1/25	4%	0.04
1/40	2 1/2%	0.025
1/50	2%	0.02

58.4 Powers of 2 and unit-digit cycles

n	2^n	3^n	5^n
1	2	3	5
2	4	9	25
3	8	27	125
4	16	81	625
5	32	243	3,125
6	64	729	15,625
7	128	2,187	78,125
8	256	6,561	3,90,625
9	512	19,683	
10	1,024	59,049	
11	2,048		
12	4,096		

Unit digit	Last digits of powers 1, 2, 3, 4 …	Cycle length
0	0	1
1	1	1
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
4	4, 6	2
5	5	1
6	6	1
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4
9	9, 1	2

58.5 Square and cube root endings

Number ends in	Square ends in	Cube ends in
0	0	0
1	1	1
2	4	8
3	9	7
4	6	4
5	5	5
6	6	6
7	9	3
8	4	2
9	1	9

58.6 Primes and divisibility

Primes up to 100									
2	3	5	7	11	13	17	19	23	29
31	37	41	43	47	53	59	61	67	71
73	79	83	89	97					

Divisor	Rule	Example
2	Last digit even (0, 2, 4, 6, 8)	3,574 ✓
3	Digit sum divisible by 3	4,512 → 12 ✓
4	Last two digits divisible by 4	7,316 → 16 ✓
5	Last digit 0 or 5	8,435 ✓
6	Divisible by 2 and by 3	2,154 ✓
7	Double the last digit, subtract from the rest; repeat	203 → 20 − 6 = 14 ✓
8	Last three digits divisible by 8	51,128 → 128 ✓

Divisor	Rule	Example
9	Digit sum divisible by 9	6,723 → 18 ✓
10	Last digit 0	4,560 ✓
11	Alternating digit sum divisible by 11	9,537 → (9+3) – (5+7) = 0 ✓
12	Divisible by 3 and by 4	1,524 ✓
13	Add 4 × last digit to the rest; repeat	247 → 24 + 28 = 52 ✓
25	Last two digits 00, 25, 50, 75	3,475 ✓

58.7 Algebraic identities

Identities

$(a + b)^2 = a^2 + 2ab + b^2$ · $(a - b)^2 = a^2 - 2ab + b^2$ · $a^2 - b^2 = (a + b)(a - b)$ $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ · $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ · $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ $(x + a)(x + b) = x^2 + (a + b)x + ab$ Quadratic roots: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$

58.8 Aptitude formulas

Percentages, profit and interest

Successive change: $a + b + ab/100$ · r% more → $r/(100 + r)$ less SP = $CP(100 + P\%)/100$ · SP = $MP(100 - D\%)/100$ · Profit % on CP SI = $PRT/100$ · $A = P(1 + R/100)^T$ · 2-year CI – SI = $P(R/100)^2$

Ratio, averages and mixtures

Share = total × part/(sum of parts) · Weighted average = $\Sigma n_i a_i / \Sigma n_i$ Alligation: cheaper : dearer = (dearer – mean) : (mean – cheaper) Replacement: $x(1 - y/x)^n$ · Partnership: capital × time

Time, work, speed

Work: total = LCM; outlets negative · $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ D = S × T · km/h × 5/18 = m/s · Average speed (equal distances) = $2ab/(a + b)$ Trains: add lengths; relative speed adds (opposite) or subtracts (same) Boats: boat = (down + up)/2, stream = (down – up)/2

Counting and probability

${}^nP_r = n!/(n - r)!$ · ${}^nC_r = n!/(r!(n - r)!)$ · P(not A) = 1 – P(A) P(A or B) = P(A) + P(B) – P(A and B)

Mensuration

Rectangle lb, 2(l + b) · Triangle ½bh, Heron $\sqrt{(s(s-a)(s-b)(s-c))}$ · Circle πr^2, $2\pi r$ Cuboid lbh, 2(lb + bh + hl) · Cube a^3, $6a^2$ · Cylinder $\pi r^2 h$, $2\pi rh$ · Cone $\frac{1}{3}\pi r^2 h$, πrl · Sphere 4/3 πr^3, $4\pi r^2$

Calendar

Odd days: ordinary year 1, leap year 2; 100 yrs 5, 200 yrs 3, 300 yrs 1, 400 yrs 0 Months: Jan 3, Feb 0/1, Mar 3, Apr 2, May 3, Jun 2, Jul 3, Aug 3, Sep 2, Oct 3, Nov 2, Dec 3

Chapter 59: Common Mistakes, Exam-Day Strategy and Checklists

59.1 Twenty common mistakes

#	Mistake	Correct idea
1	Forgetting to carry in Urdhva	carry everything except one digit from each column
2	Wrong number of digits on the right in Nikhilam	right part has as many digits as the base has zeros
3	Sign errors with mixed deviations	one above and one below → right part is negative
4	"All from 9" applied to the last digit	the last non-zero digit is from 10
5	Ekanyunena with unequal digit counts	pad with zeros or use a base method
6	Squares ending in 5 for 3-digit numbers	$n(n + 1)$ still works: $115^2 \rightarrow 11 \times 12 \mid 25$
7	Treating $a^2 + b^2$ as $(a + b)^2$	remember $2ab$
8	Profit % on SP	profit % is on CP
9	Averaging speeds	use $2ab/(a + b)$
10	Adding successive percentages	$a + b + ab/100$
11	Wrong units (km/h with metres)	convert first
12	Leap-year rule for 1900	1900 is not a leap year
13	Digit-sum check "proves" an answer	it only rejects wrong answers
14	Over-rounding in approximation	round to the nearest easy number
15	Missing the overlap in "or" probability	subtract P(A and B)
16	Remainder taken as negative	add the divisor to make it positive
17	$-(a - b)$ written as $-a - b$	$-(a - b) = -a + b$
18	Misreading the scale in DI	read title, units and scale first
19	Using a shortcut that does not fit	check the condition (same tens? near a base?)
20	Spending too long on one question	use the two-pass method

59.2 Exam-day strategy

1. Sleep well the night before. A tired brain is slower than any shortcut can make up for.
2. Warm up for 5 minutes: a few squares, tables and percentages.
3. Read each question fully before calculating.
4. First pass: easy questions only. Second pass: marked questions.
5. Estimate before choosing; check the units digit and the size of the answer.
6. With negative marking, avoid pure guesses. Guess only after eliminating at least two options.

59.3 Revision checklists

One week before the exam

- Revise all recap boxes.
- Take 3 full mocks and analyse each one.
- Re-do all circled mistakes.

One day before the exam

- Read this chapter's formula sheets once.
- Do one light timed test. Do not start anything new.

Skills checklist

- [ ] Squares to 30 and cubes to 15 from memory
- [ ] Fraction–percentage chart from memory
- [ ] Nikhilam and Urdhva without writing intermediate steps
- [ ] Divide by 9 and by numbers near 10/100
- [ ] All aptitude formulas in this chapter
- [ ] Two-pass method practised in at least 5 mocks

> **Final words**
>
> Vedic Maths is a toolbox, not magic. Understand why each method works, choose the right tool for each problem, check your answers, and practise a little every day. Good luck on your journey from zero to hero!